

NHGRI AnVIL

Project Description

The NHGRI AnVIL is a cloud based environment that hosts high value data sets and commonly used bioinformatics tools in a secure FedRAMP-certified environment that can scale to meet the computational needs of researchers. The AnVIL will provide automated data access with the DUOS platform. AnVIL users have access to extensive training materials spanning from general genomics to advanced analysis methods using modular and open access Massively Open Online Courses (MOOCs).

Participants

Funder: National Human Genome Research Institute, Eric Green (director), Valentina Di Francesco (program officer), Ken Wiley (program officer)

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Institutions: The Broad Institute, Johns Hopkins University, University of Chicago, Penn State University, University of California, Santa Cruz, Oregon Health and Sciences University, Harvard University, Vanderbilt University, Roswell Park Comprehensive Cancer Center, Washington University, City University of New York, American Heart Association, Carnegie Inst. for Science, Yale University

Data available

AnVIL hosts CCDG, CMG, GTEx, and 1000 Genomes. As data sets are added to these groups they will become available on AnVIL. EMERGE will become available on AnVIL in 2022.

Source	Cohorts	Samples	Participants	Size (TB)
CCDG	198	270,135	256,318	2,582
CMG	39	15,732	14,973	61
1000 Genomes	1	3,202	3,202	73
GTEx (v8)	1	17,382	979	182
Convergent Neuro	2	304	304	5
HPRC	1	57	47	160
PAGE	4	690	690	17
WGSPD1	5	1,504	9,943	176
T2T	1	0	3219	503
eMERGE*	Pending	Pending	Pending	Pending
All Datasets	252	309,006	289,675	3,759

Tools available

AnVIL uses the Terra platform to launch and run tools on Google Cloud Platform within a FedRAMP-certified secure boundary.

Users can currently run batch analysis with WDL and interactive analysis with Bioconductor, Galaxy, seqr and Jupyter Notebooks supporting Python, R and RStudio. The Dockstore workflow repository is integrated with Terra, providing access to hundreds of published workflows.

AnVIL enables users to bring their own tools to the platform.



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Authentication

Both Google emails and NIH RAS IDs are used as an authentication mechanism for controlled access data

Authorization

Consortium and developer whitelists are maintained to provide access to data

General users submit DARs through dbGAP

Indexing

Data objects are assigned permanent globally unique IDs (GUIDs) to allow for access across tools, without requiring copies be created and transferred

Datasets are identified through faceted search over phenotypic data

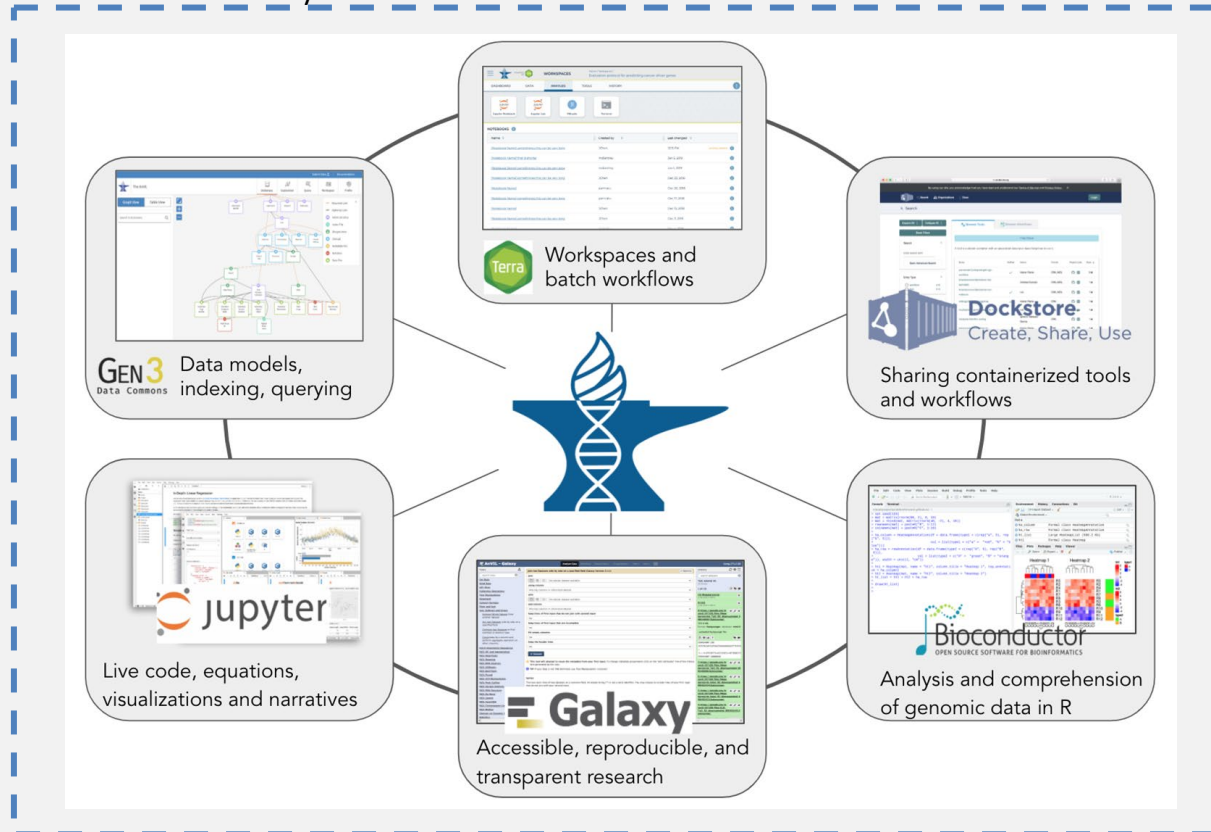
Data Models

AnVIL has adopted the Terra Interoperability data model, an expandable data mode to support multiple, diverse data models.

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Architectural Diagram

All data use and analysis in a FedRAMP-certified environment



FedRAMP certified
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Implemented on

Google Cloud Platform